

Letter : Sumit Jadhav

Subject: Technical Assessment – AI/ML Engineer (Computer Vision)

Hi Sumit,

Thank you for your interest in [Company Name]. As the next step in our hiring process, please complete the assessment below within 72 hours of receiving this email.

Task: Helmet Safety Violation Detector – Train, Optimize, Deploy

Build a system that detects workers without helmets in video and reports each violation once.

Part 1 – Training

- Fine-tune YOLOv8 (n or s) on a public helmet/PPE dataset of your choice (e.g., from Roboflow Universe) with at least two classes: helmet, no-helmet.
- Report mAP@50 and mAP@50-95 per class on a held-out test set.
- Show 5 examples where your model fails and explain why.

Part 2 – Optimization

- Export to ONNX, then create a quantized version (INT8 or FP16).
- Benchmark on CPU: PyTorch vs ONNX vs quantized ONNX. Report FPS, ms per frame, model size (MB), and mAP for each.
- Explain how much accuracy you lost from quantization and whether the trade-off is worth it.

Part 3 – Video Violation Tracking

- Process the provided video using object tracking (e.g., ByteTrack).
- Count UNIQUE violations: a worker without a helmet should be counted once, not once per frame.
- A worker who removes their helmet partway through the video counts as a violation from that moment.
- Save a snapshot image of each unique violation with a timestamp.

Part 4 – API

- FastAPI endpoint: upload a video → returns a JSON report (total workers, unique violations, timestamps, snapshot paths).
- Processing must not block the server for other requests.

Input video: [paste video link here]

Submission

- GitHub repository link
- Output video with boxes, track IDs and violation alerts (Google Drive link)
- README with: setup steps, training details, benchmark table (include your CPU model), results, and a section called “What didn’t work / what I’d improve”
- A 3–5 minute video (camera on) explaining your results and the hardest problem you faced. Upload it to Google Drive or Loom and share the link.

GitHub link: _____

Output video link: _____

Explanation video link: _____

Note: AI tools are allowed. Your system will also be tested on a different video.

If you have any questions about the task, reply to this email.

Best regards,

[Your Name]

[Company Name]